HARD Problem

Title

Kaprekar's Constant (6174) Transformation Count

Topics

Number Manipulation, Digit Extraction, Sorting Logic (Manual), Loops, Conditional Structures, Mathematical Simulation

Difficulty Level

Hard

Problem Detailed Statement (Scenario Based)

In an advanced mathematical programming workshop, students are introduced to a fascinating numerical process known as **Kaprekar's Routine**.

For any 4-digit number (with at least two distinct digits), the following process is applied:

1. Arrange its digits in **descending order** to form the largest number.
2. Arrange its digits in **ascending order** to form the smallest number.
3. Subtract the smaller number from the larger number.
4. Repeat the process with the result.
5. Continue until the number becomes **6174** (Kaprekar's Constant).

It is mathematically proven that for most valid 4-digit numbers, this process eventually converges to **6174** within a maximum of 7 iterations.

Example:

Let N = 3524

Step 1:
Descending → 5432
Ascending → 2345
5432 − 2345 = 3087

Step 2:
8730 − 0378 = 8352

Step 3:
8532 − 2358 = 6174

Process stops.

Total steps = 3

Now, students are given a 4-digit integer **N**.

They must:

- Validate that N is a 4-digit number.
- Ensure it does not contain all identical digits (like 1111).
- Implement digit extraction manually.
- Sort digits without using built-in sorting functions.
- Repeat Kaprekar transformation.
- Count how many iterations are required to reach 6174.
- If the number never reaches 6174 (invalid case), print -1.

Restrictions:

- Do not convert number into string.
- Do not use built-in sorting utilities.
- Use loops and conditional logic only.
- Digit sorting must be done manually (using nested loops or comparison logic).

This problem tests:

- Advanced number manipulation
- Nested loop logic
- Mathematical simulation
- State repetition handling
- Loop termination control
- Deep logical reasoning

Input Format

A single integer:

`N`

Constraints:

`1000 ≤ N ≤ 9999`

Additional Rule:

- N must have at least two distinct digits.

Output Format

- Print the number of iterations required to reach 6174.
- If invalid input or all digits same, print:

```
-1
```

Examples

Example 1

Input:

```
3524
```

Output:

```
3
```

Example 2

Input:

```
6174
```

Output:

```
0
```

Example 3

Input:

```
1111
```

Output:

```
-1
```

Example 4

Input:

```
9831
```

Output: